

Hydraulic power unit - design brief

AI BRAKING / AB-HPU / C0 / 2026-09-07

A compact reservoir and removable equipment deck reserve space for a bought motor-pump, valve manifold and filtration. This is packaging CAD, not a designed hydraulic circuit.

TARGETS (not validated)

Reservoir outside envelope: 400 × 260 × 220 mm. Geometry

Gross internal volume: ≈21.5 L. Unbaffled volume; usable oil level unassigned

System pressure / flow: Unassigned. Derive from selected caliper and release timing

Circuit architecture: Pending safety analysis. No pressure-retaining custom manifold design

DESIGN DECISIONS

Buy qualified hydraulic components and manifold rather than machine custom pressure passages.

Use a removable deck for bench service and clearly separated fill/drain access.

Allocate pressure sensing and filtered breather space early; select hardware after the circuit review.

CANDIDATE MATERIALS

Reservoir: S355J2+N candidate welded sheet; cleanliness and coating compatibility require review.

Pump, motor, valve block and filter: purchased certified components.

Assembly and engineering gates

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ASSEMBLY INTENT

- 1. Approve circuit and equipment list before cutting sheet.
- 2. Fabricate and leak-test reservoir to approved welding and cleaning procedure.
- 3. Install removable deck and selected component brackets.
- 4. Fit piping, flushing, relief protection and controls per approved hydraulic schematic.

OPEN DESIGN RISKS

- 1. No circuit, relief setting, accumulator sizing or failure-state analysis is completed.
- 2. Reserved component volumes are not vendor dimensions.
- 3. Reservoir baffles, suction conditions and thermal dissipation remain unresolved.

VALIDATION REQUIRED

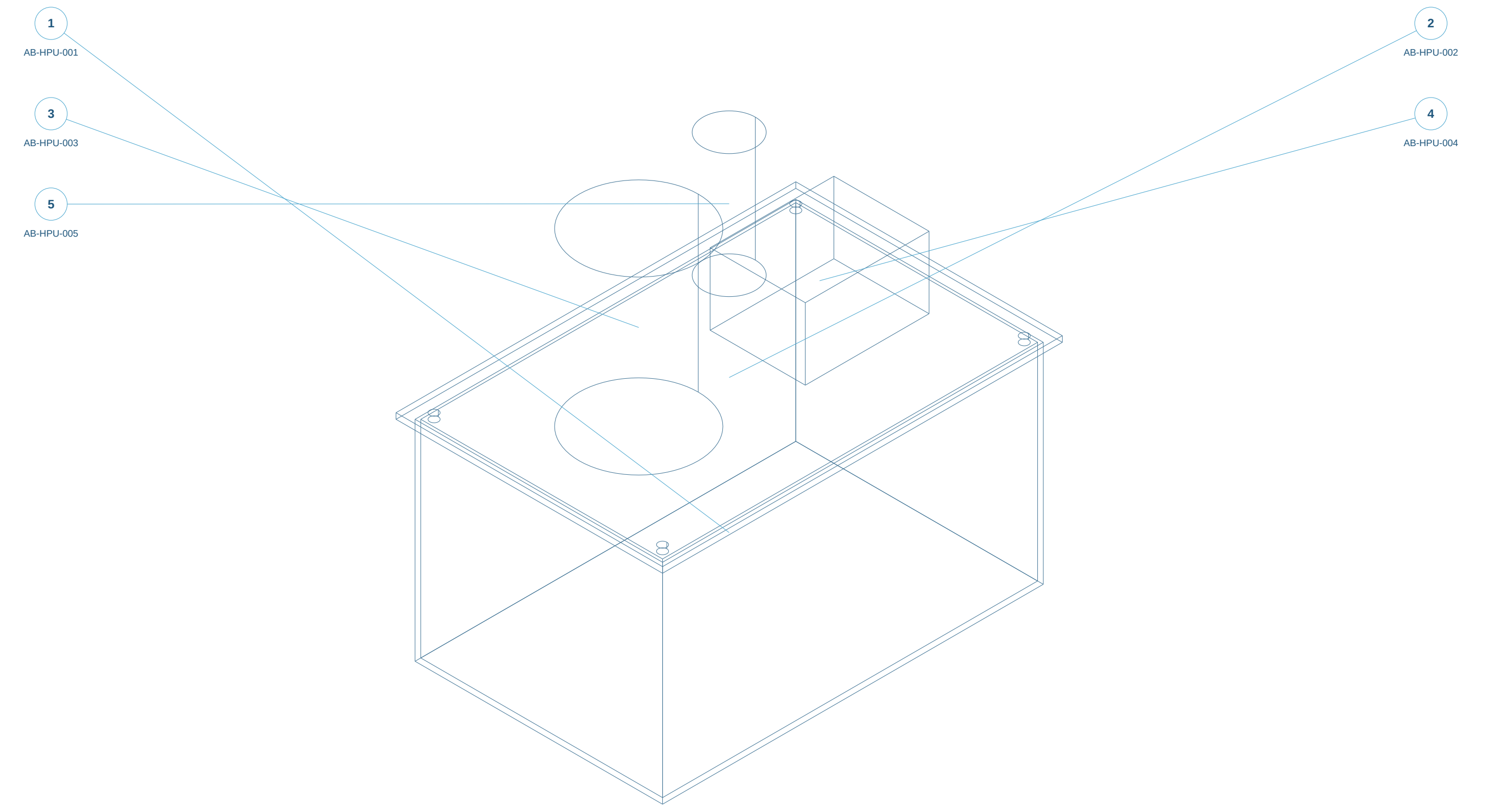
- 1. Circuit failure analysis, pressure transients, oil temperature and cleanliness.
- 2. Leak, flush, loss-of-power and repeated application/release tests.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

- 1. Define pressure transducer, oil level/temperature, filter clog switch, motor starter, E-stop interaction and independent safe valve state.

Assembly reference / numbered components

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Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-HPU-001 | Reservoir shell | make-concept

S355J2+N candidate; condition and design properties unapproved; Fabricate welded sheet; clean and leak-test after approval

AB-HPU-002 | Removable deck | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-HPU-003 | Motor-pump envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy selected motor-pump

AB-HPU-004 | Qualified valve block envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy manifold; no custom pressure channels

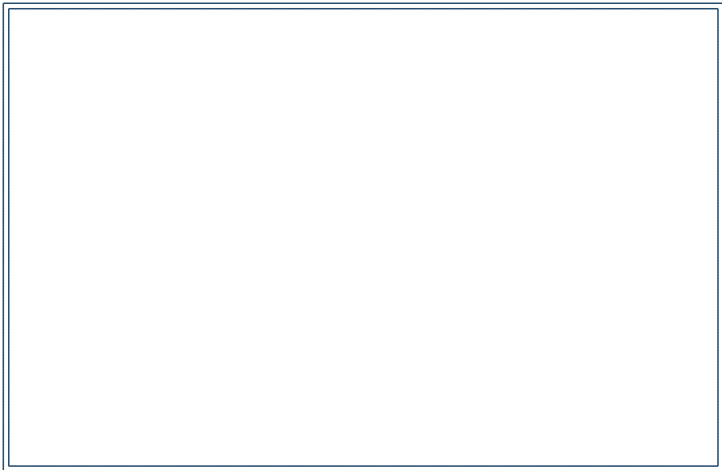
AB-HPU-005 | Filter envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy selected filter

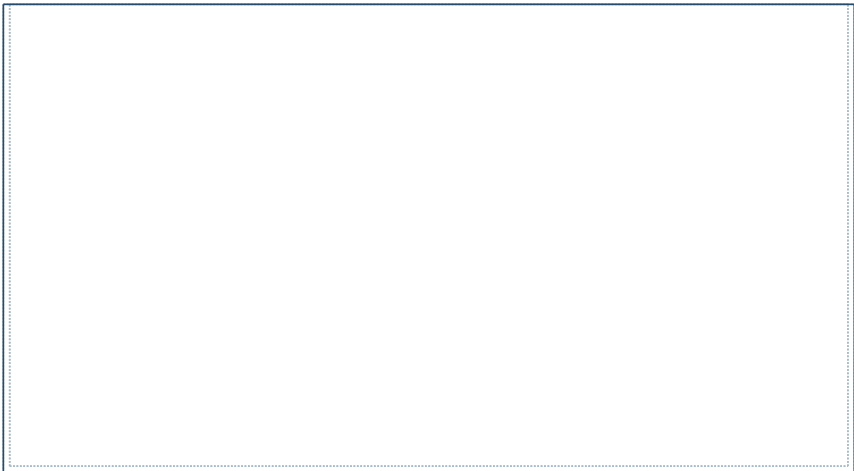
AB-HPU-001 / Reservoir shell

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Fabricate welded sheet; clean and leak-test after approval

Overall X/Y/Z bounds: 400.00 / 260.00 / 220.00 mm

Outer 400 x 260 x 220; nominal wall and floor 3 mm.

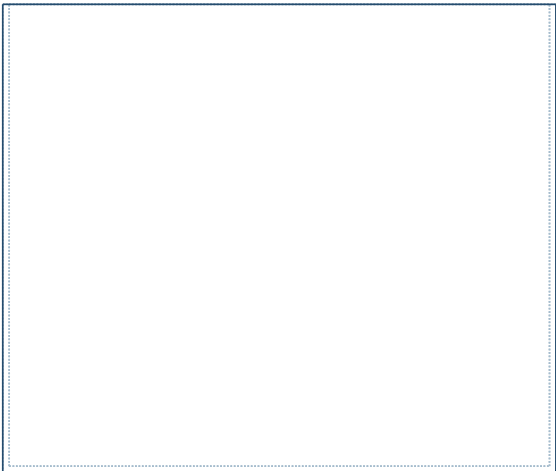
No baffles, drains or weld details released; open top.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



AB-HPU-002 / Removable deck

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 420.00 / 280.00 / 6.00 mm

420 x 280 x 6; four diameter 9 holes at (+/-190,+/-120).

Tank mating flange/fasteners and gasket pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

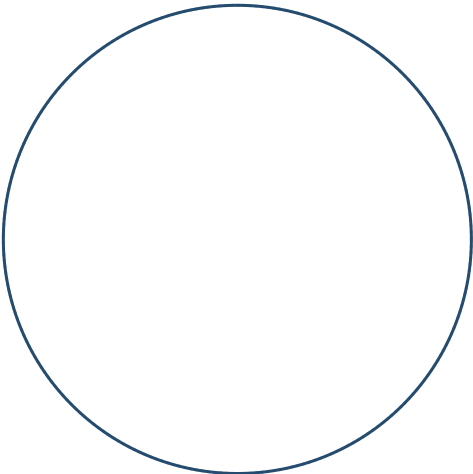
SIDE Y-Z



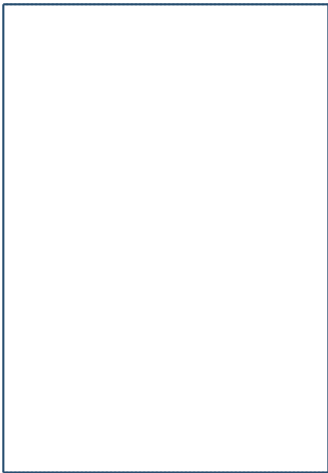
AB-HPU-003 / Motor-pump envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy selected motor-pump

Overall X/Y/Z bounds: 125.00 / 125.00 / 180.00 mm

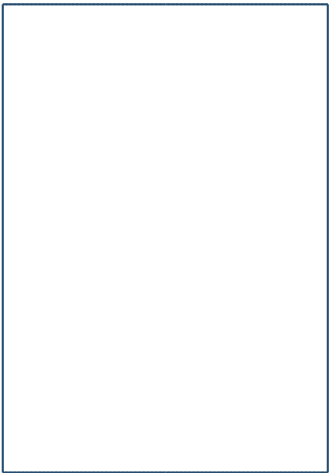
All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



AB-HPU-004 / Qualified valve block envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy manifold; no custom pressure channels

Overall X/Y/Z bounds: 130.00 / 100.00 / 75.00 mm

All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

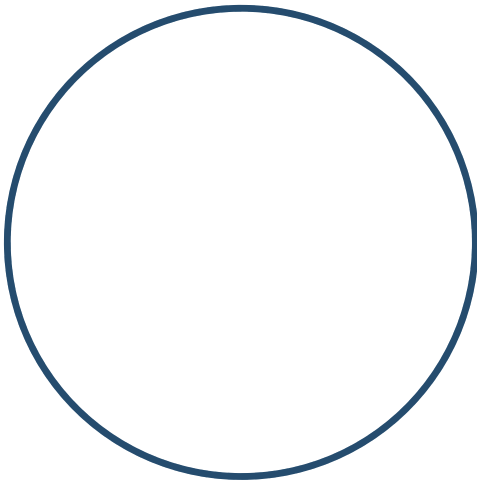
SIDE Y-Z



AB-HPU-005 / Filter envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy selected filter

Overall X/Y/Z bounds: 55.00 / 55.00 / 130.00 mm

All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z

